HACKATHON PROJECT

Sonar: Rock or Mine?

Classifying underwater sonar signals as rocks or mines using machine learning

1. Problem Statement

Sonar systems send sound pulses underwater and record the echoes that bounce back. A metal mine and a similarly shaped rock can return very different echo patterns, but telling them apart by eye is slow and error-prone. The challenge is to build a system that takes the readings of a sonar signal and predicts whether the object is a **rock (R)** or a **mine (M)**.

2. Objectives

- Train a machine learning model on labelled sonar data to separate rocks from mines.
- Build a simple web interface where anyone can upload a dataset and preview it.
- Let a user paste a single sonar reading and instantly get a Rock or Mine prediction with a confidence score.
- Keep the project easy to set up and run from the GitHub repository.

3. Dataset

The project uses the classic sonar dataset: each row is one sonar signal, described by 60 numeric values followed by a class label.

Item	Description
Rows	One sonar signal (echo) per row
Features	60 numbers, each the energy in a frequency band of the returned signal (range 0.0 to 1.0)
Label	R = Rock, M = Mine
File format	CSV, no header row, kept next to app.py

4. Approach

- **Load** the CSV with pandas and separate the 60 features from the label.
- **Split** the data into training and test sets to measure how well the model generalises.
- **Train** a classifier with scikit-learn on the training data.
- **Evaluate** accuracy on the unseen test set.
- **Serve** the model through a Flask API (`/predict`) that returns the class and a confidence value.

5. Web Application

The front end is a lightweight HTML, CSS and JavaScript page that talks to the Python server.

Feature	What it does
Upload datasheet	Choose a CSV file and preview the first 100 rows in a scrollable table
Copy row	One-click button to copy any previewed row to the clipboard
Paste a row	Paste a tab-separated row into the box and submit it
Prediction	Shows Rock or Mine with a confidence percentage
About	Project tabs with the GitHub repository, contributors and related files

6. Tech Stack

Layer	Tools
Front end	HTML, CSS, JavaScript
Back end	Python, Flask
Machine learning	pandas, NumPy, scikit-learn

7. How to Run

- Download and install Python from python.org (tick *Add python.exe to PATH* on Windows).
- Clone or download the project and open the folder in a terminal.
- Install the modules:

```
pip install pandas numpy scikit-learn flask
```

- Make sure the sonar CSV is in the same folder as app.py, with the filename the code expects.
- Start the server:

```
python app.py
```

- Open the localhost link printed in the terminal (usually http://127.0.0.1:5000).
- Upload the datasheet, copy a row, paste it in the box and submit.
- The app shows whether the signal is a **Rock** or a **Mine**.

8. Team and Repository

Contributor	GitHub
Rahul Singh	github.com/RahulSingh2008
Shivam	github.com/officegitshivam

Repository: github.com/RahulSingh2008/sonar-signal-classifier

9. Future Improvements

- Compare several models (for example Logistic Regression, SVM, Random Forest) and pick the best.
- Show feature importance to explain why a signal was classified as a rock or a mine.
- Accept row uploads directly from the table without copy and paste.
- Deploy the app online so no local setup is needed.